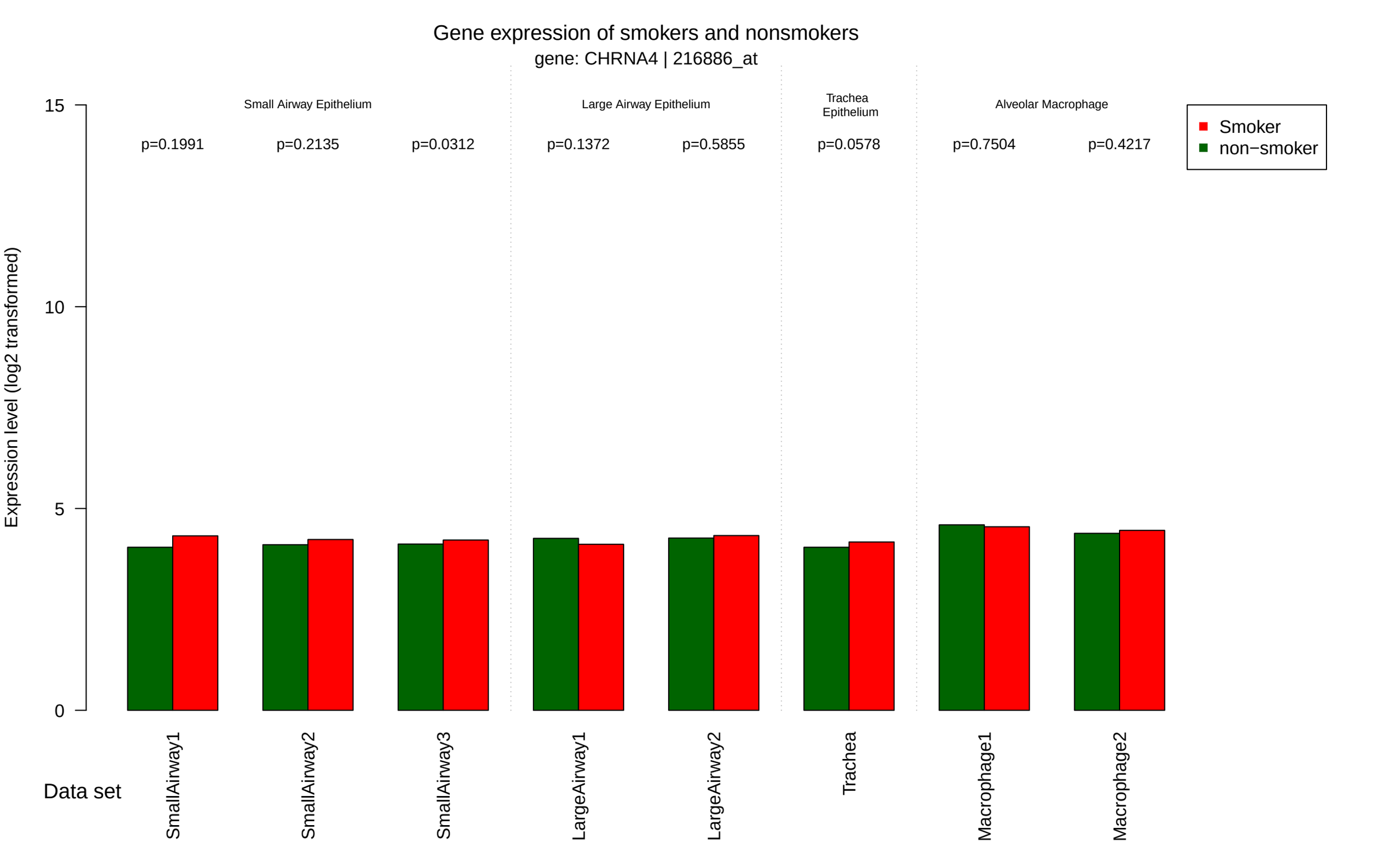
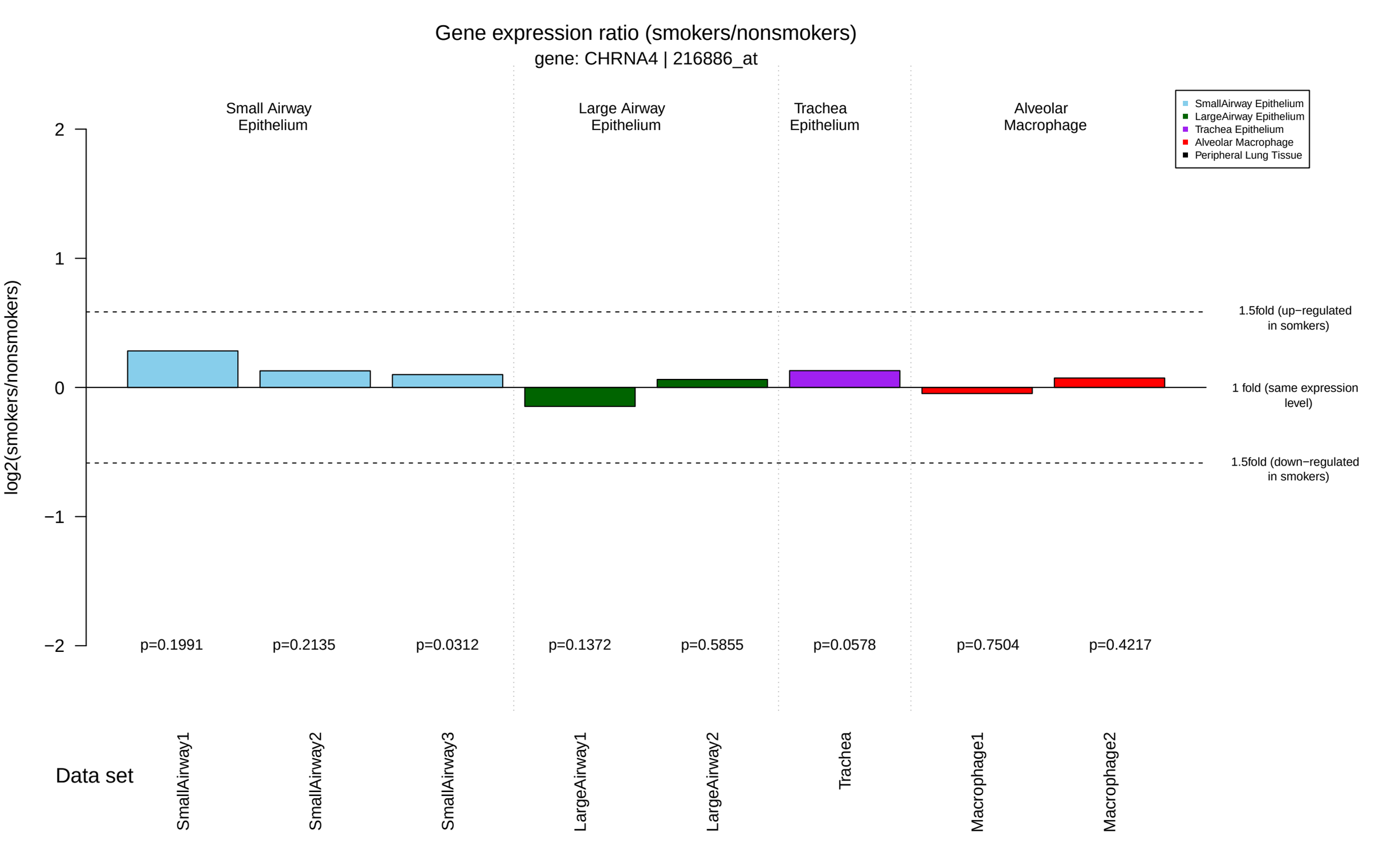
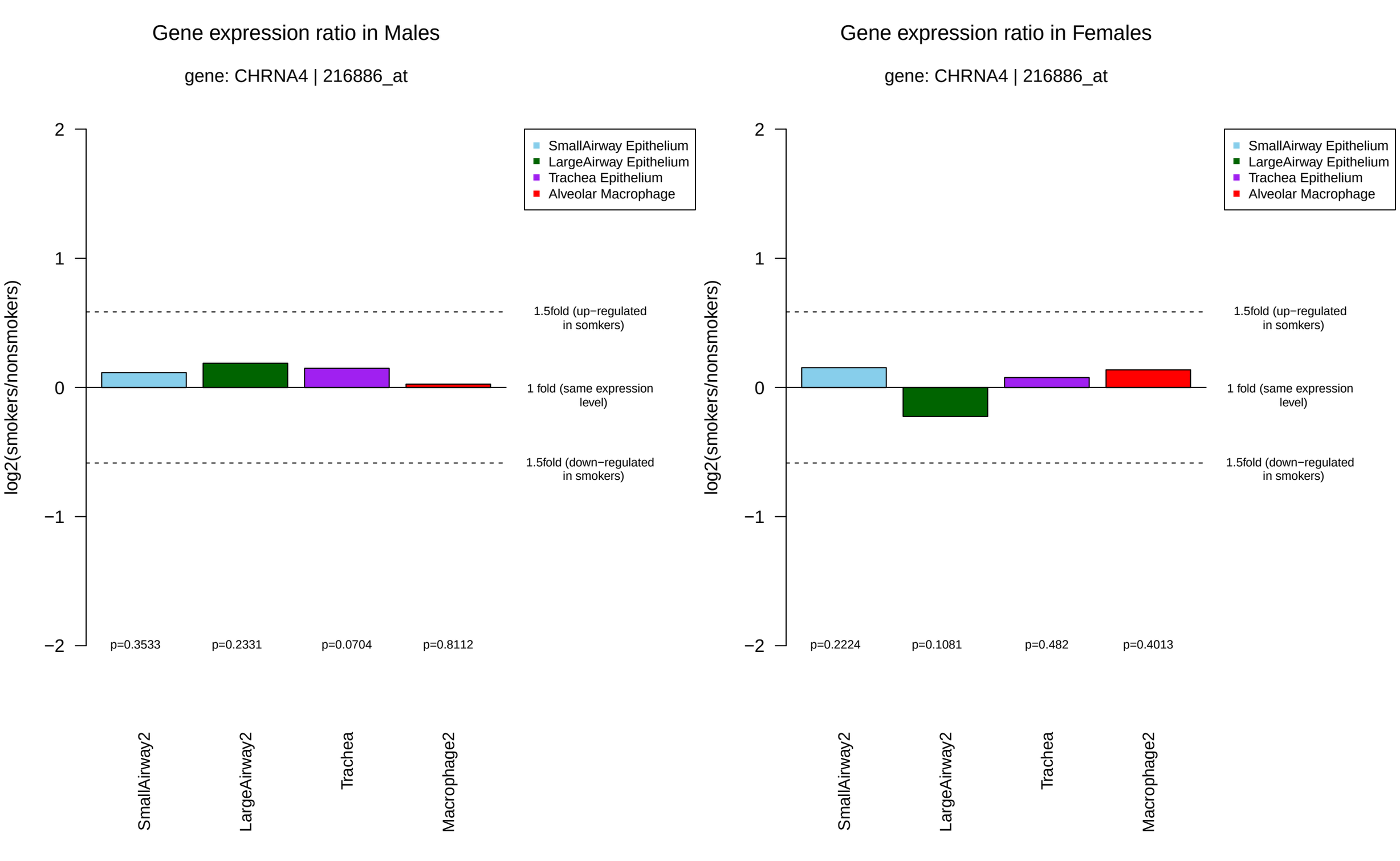
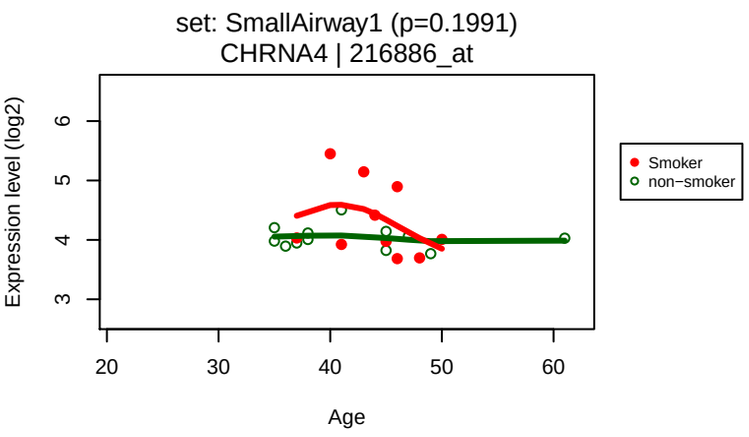
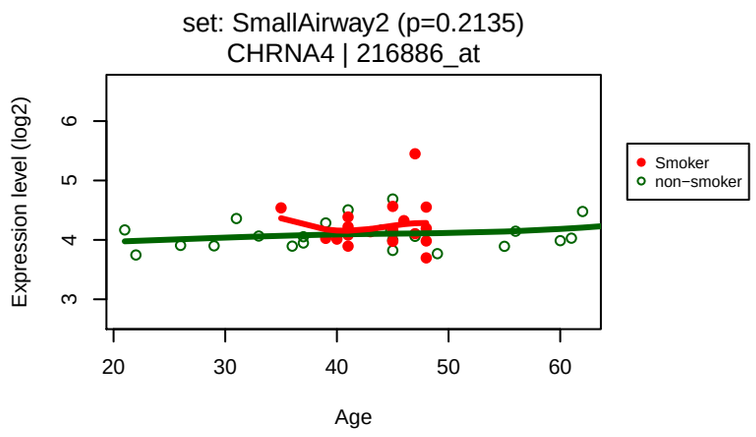
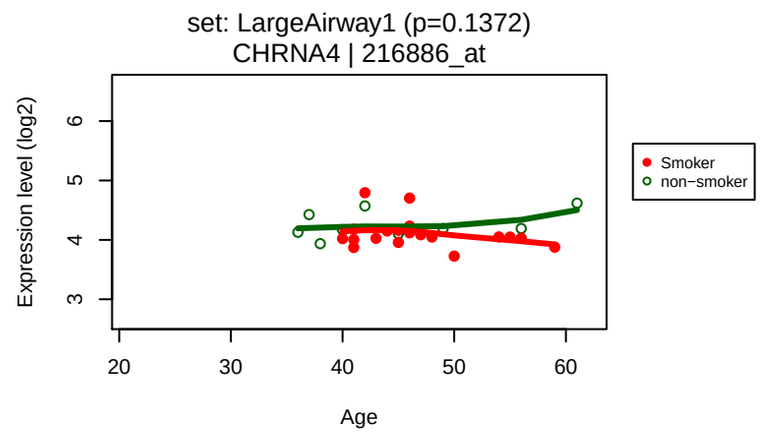

Gene expression of smokers and nonsmokers
gene: CHRNA4 | 216886_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1991
p=0.2135
p=0.0312
p=0.1372
p=0.5855
p=0.0578
p=0.7504
p=0.4217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

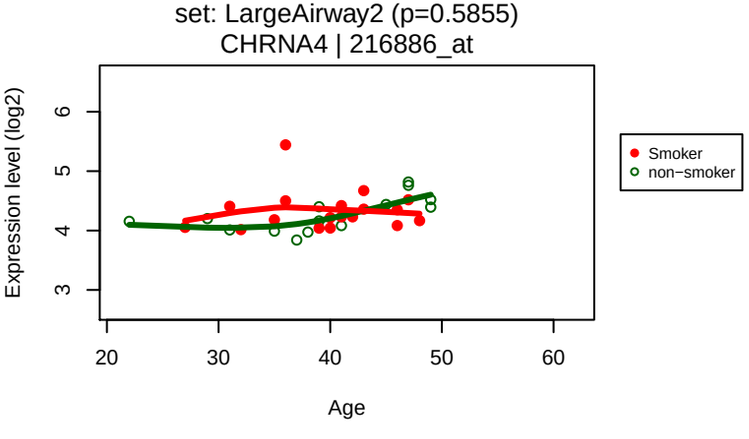

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA4 | 216886_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1991
p=0.2135
p=0.0312
p=0.1372
p=0.5855
p=0.0578
p=0.7504
p=0.4217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

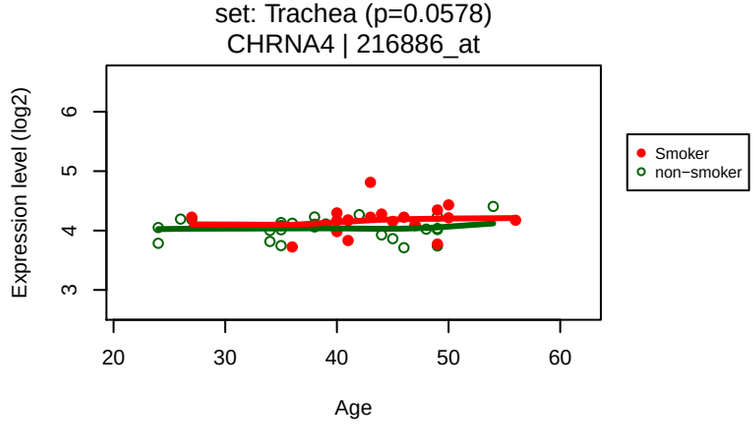

Gene expression ratio in Males
gene: CHRNA4 | 216886_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3533
p=0.2331
p=0.0704
p=0.8112
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA4 | 216886_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2224
p=0.1081
p=0.482
p=0.4013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

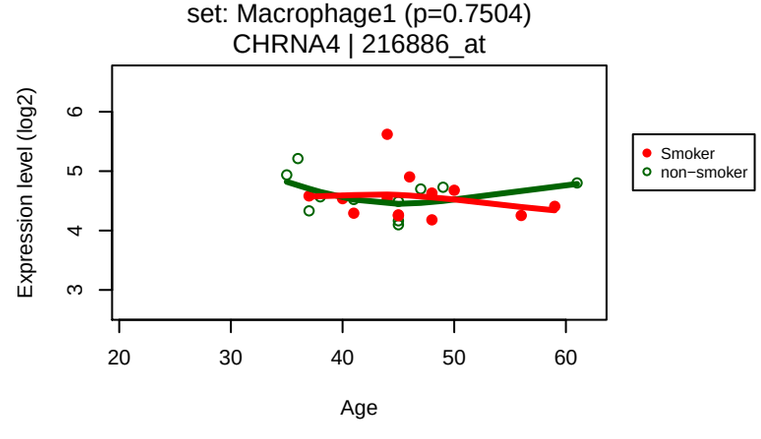

set: SmallAirway1 (p=0.1991)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker

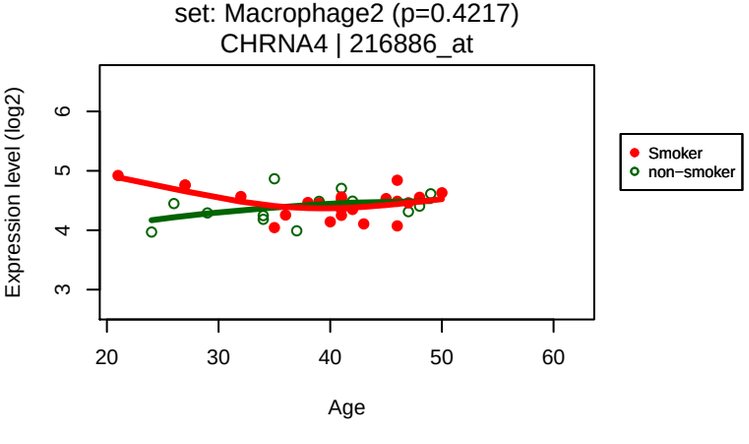

set: SmallAirway2 (p=0.2135)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker

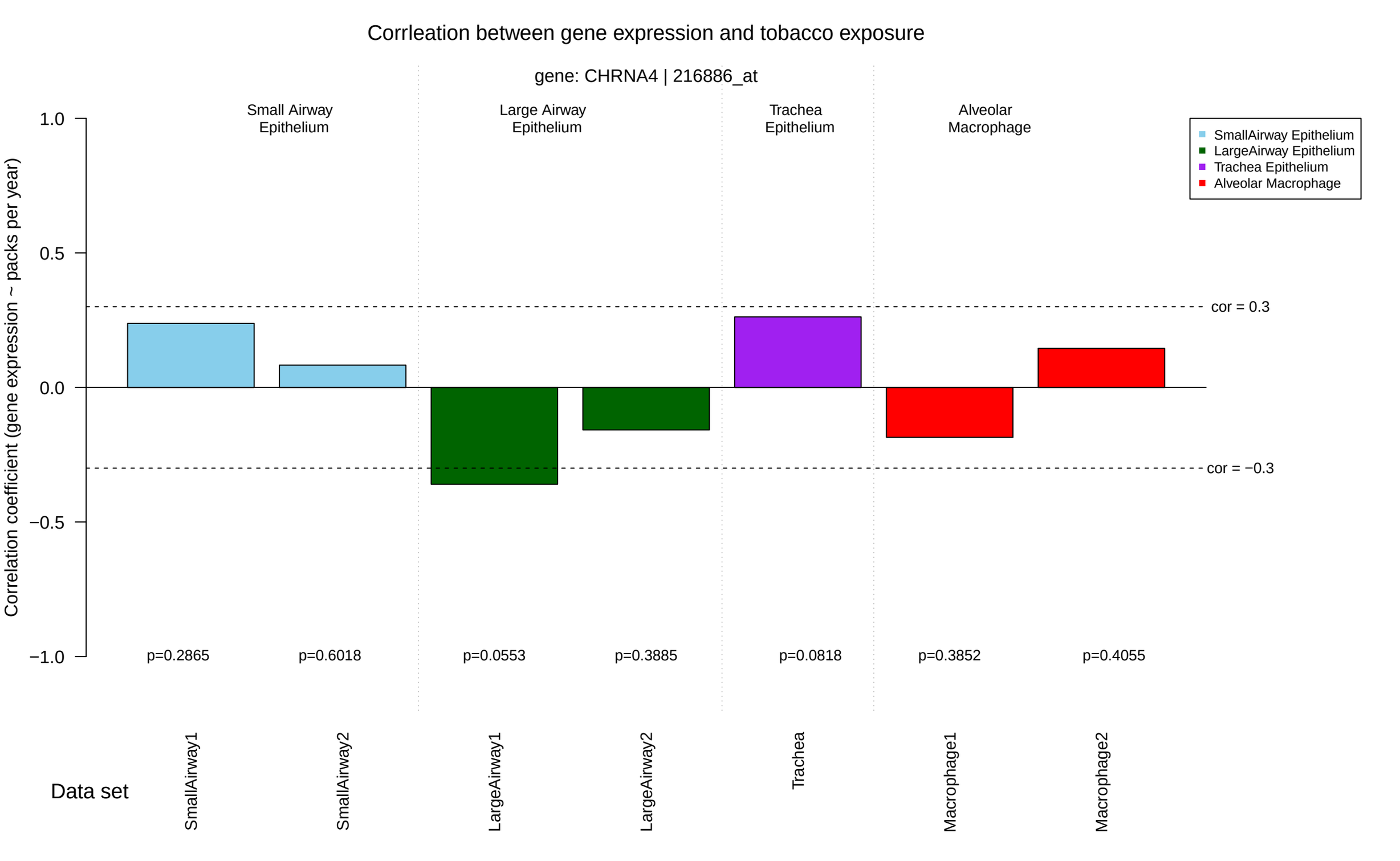

set: LargeAirway1 (p=0.1372)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5855)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0578)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7504)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4217)
CHRNA4 | 216886_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA4 | 216886_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2865
p=0.6018
p=0.0553
p=0.3885
p=0.0818
p=0.3852
p=0.4055
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2